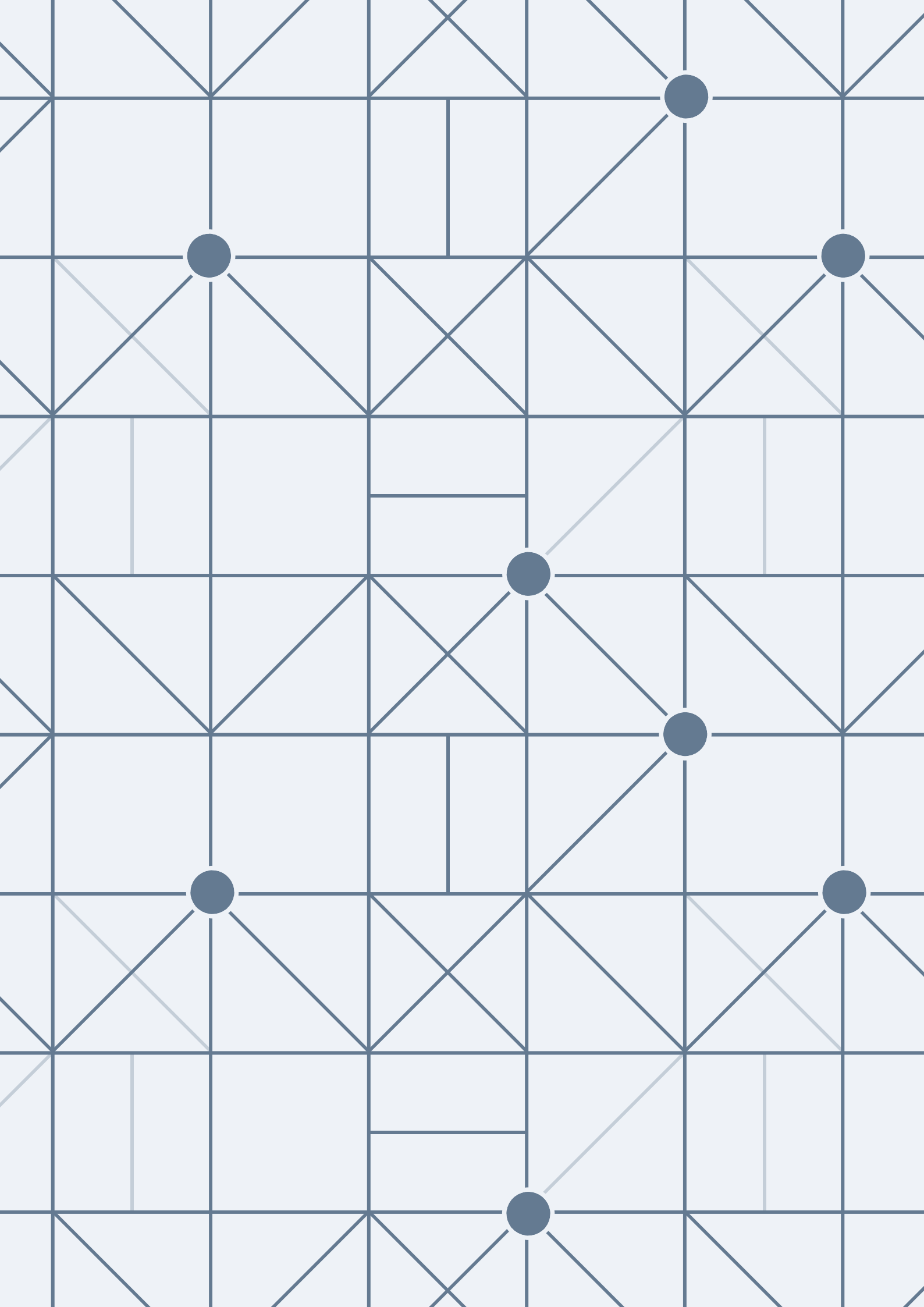




ideal gases

ideal gas $\Rightarrow$ low pressure $\Rightarrow$ $N \ll N_0$

in compressed matter

$$p(z) = \frac{g^{(N)}(z)}{Z} = \frac{e^{-\beta E(z)}}{Z} \Rightarrow \text{probability of state } z \text{ due to } p(z) \text{ but in actual state } z \text{ or } p(z)$$

If N low, a particle $N(z) \Rightarrow 1 + g(z)$ but if N low, a particle $N(z)$ due $N = \sum_i g_i = \sum_i N_i$
 actually particle $N(z)$ due $N = \sum_i g_i = \sum_i N_i$
 high $\Rightarrow$ many particles

for example for (very small) $N = \sum_i g_i = \sum_i (1 + g_i) = N_0 + N_1$ at g_1

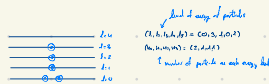
is not with accuracy

$$N_1 = 2g_1 = 2g_1$$

is not correct with accuracy

Difference for

N is number of particles and boxes



Grand module

$$Z = \sum_{\{n_i\}} e^{-\beta E(\{n_i\})} \Rightarrow \text{in } g_i \text{ states of } i \text{ box of } i \text{ box} \Rightarrow \sum_i g_i e^{-\beta E_i} \Rightarrow \sum_i g_i e^{-\beta E_i}$$

the particles are distinguishable
 consider all particles the same
 then we must have law to
 distribute the particles in g_i energy levels

$$\Rightarrow \text{A set of } \{g_i\} \text{ occupied } \frac{g_i!}{N!} \text{ particles values of } \{E_1, E_2, \dots, E_N\} \Rightarrow Z = \sum_{\{n_i\}} \frac{g_i!}{N!} e^{-\beta E_i}$$

Maxwell-Boltzmann approximation: $N \gg g_i$

$$g_i \gg 1 \Rightarrow \text{the approximation of good value TTT or } g_i \gg 1$$

$$N_i = \frac{1}{e^{\beta E_i} - 1}$$

Under these conditions, laws and laws have the same way

with Maxwell-Boltzmann approximation, we get the law

$$Z = \sum_i g_i e^{-\beta E_i} \Rightarrow Z = \sum_i g_i e^{-\beta E_i} \Rightarrow \sum_i g_i e^{-\beta E_i}$$

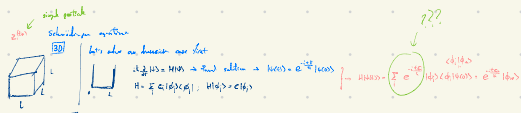
$E = E_1, E_2, \dots, E_N$

$$Z = \sum_i g_i e^{-\beta E_i} \Rightarrow Z = \sum_i g_i e^{-\beta E_i} \Rightarrow \sum_i g_i e^{-\beta E_i}$$

$Z = \sum_i g_i e^{-\beta E_i}$

with $N \gg g_i$
 actual law of particles
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